

Control and Regulation Systems

Physical Measuring Techniques FK7063

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Introduction

In this laboratory exercise, a PID control system will be used to control a pump with the goal of keeping a constant water level in a small tank. The control system has three tweakable parameters which will be combined in different ways to optimize the pump's behaviour.

1 Theory and Background

Control systems are used to regulate the behaviour of a system (the *controlled system*) that depends on various variables. In the case of this experiment, the controlled system is a pump leading into a water tank whose water level is measured by a sensor. The control system is a microcontroller (Arduino) that receives information about the water level and can regulate the speed of the pump.

PID controllers are a type of control system that regulates control variables (the speed of the pump) based on the difference between the controlled variable (the water level in the tank) and a reference value (the desired water level). This difference is called the error $e(t)$ and a PID system regulates the control variable according to three terms derived from the error:

- P – Proportional $\propto e(t)$
- I – Integral $\propto \int e(t)dt$
- D – Differential $\propto \frac{de}{dt}$

The strength of the individual terms can be controlled by three parameters P , I and D .

2 Setup and Experiment

The experimental setup consists of a pump leading into a water tank. The pump is controlled by an Arduino based on the water level in the tank. This works by means



Figure 1: Water level development with only the propotional (P) term. Parameters and coefficient values are shown as sliders near the top of the window. The order is: target level, maximum level, P , I , D . The left two graphs show the water level over time.

of a PID controller implemented in software. This implementation is accompanied by a graphical user interface that can be used to change parameters and watch parameter changes over time.

Different parameter combinations are tested to understand the behaviour of the PID system.

2.1 Proportional Term

We begin by only using the proportional term. One choice of parameters and the resulting development of the water level can be seen in Fig. 1. We can see that a high P parameter leads to oscillations of the water level. This is caused by the delay between a change in the Arduino's output and the resulting change in the actual water level. In the figure, the parameter is deliberately chosen such that there are stable oscillations, i.e. the oscillations do not increase or decrease in amplitude over time. For a target water level of 257, the point of stable oscillations is $P_0 = 3.94$ where the period of oscillations is roughly $T_0 = 6.2$ s.

Trying out different combinations, we see that the value of P_0 is not universal but depends on the target water level. This is because the pressure acting back on the pump is higher with a higher water level, making the same pump activation cause a lesser increase in the level, so that a higher P parameter is required for stable oscillations.

Generally, oscillations are unwanted behaviour of a control system. However, determining P_0 can be useful to find good parameter values based on a set of rules. In our case, such methods did not prove to be very effective and parameters were mostly determined by trial and error.



Figure 2: Water level development with the propotional (P) and integral (I) term.

Looking at the level around which the oscillations occur, we see that it is significantly lower than the actual target level. This is because (if we ignore the effects leading to oscillations) at some point the pressure exerted by the pump (proportional to ...) and the back-pressure due to gravity reach an equilibrium at some point.

2.2 Proportional and Integral Term

Next we add the integral term, see Fig. 2, where we have chosen $I \neq 0$ and P low enough so that there are no oscillations. Compared to only using the P term, we can see that the water level settles at the target level instead of having a permanent offset. This is because, whenever there is an offset for an extended period of time, the integral of $e(t)$ increases until the offset is zero. The Integral term thus adds a permanent amount of pump activation that is enough to put the equilibrium point at the target level instead of below it.

2.3 Proportional, Integral and Derivative Term

3 Modulation Experiment

4 Conclusion